Vivli AMR Surveillance Data Challenge Final Submission Document Cover Letter

This is the final submission document cover letter details form. Your final submission must be uploaded to your data request chat and this form completed by the specified deadline to be eligible for the next phase of the Vivli AMR Surveillance Data Challenge.

1. AMR Register Vivli ID *

Your team must have an approved expression of interest as an AMR Register Data Request. If your team has not submitted the expression of interest yet, please view the directions for participation in the Vivli AMR Surveillance Data Challenge at: https://amr.vivli.org/data-challenge/how-to-participate/
AMR Register data requests can be submitted at searchamr.vivli.org

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2. Has your team uploaded the final submission document to your team's data request chat? *

- (selected) Yes

3. We confirm that our team member form is up-to-date. *

If your team composition changed from when your team initially submitted the expression of interest. Please submit an updated team form prior to completing this cover letter form. The team form is located here: https://forms.cloud.microsoft/r/KEvT62ZUaT

- (selected) Yes - There were no changes
- Yes - We've submitted an updated form

4. Is your team entering the Vivli AMR Surveillance Data Challenge as a Student Team? *

To be eligible as a Student Team, only one member can be a non-student. That individual cannot be the lead applicant.
By entering as a Student Team, your team will eligible to win only student team awards and the Global AMR R&D Hub Cross-Domain Award (if applicable).

- ○ Yes
- ◉ No

5. Datasets included in the analysis *

Please indicate ALL datasets that were analyzed and included in the final submission document. Please also include any external datasets that were used in the analysis by name and url (if possible)

- [ ] GSK - SOAR 201818
- [ ] GSK - SOAR 201910
- [ ] GSK - SOAR 207965
- [ ] Johnson & Johnson - DREAM
- [x] Paratek - KEYSTONE
- [x] Pfizer - ATLAS_Antibiotics
- [ ] Pfizer - ATLAS_Antifungals
- [ ] Pfizer - SPIDAAR
- [ ] VenatoRX - GEARS
- [ ] Venus Remedies - PLEA I
- [ ] Venus Remedies - PLEA II
- [ ] Venus Remedies - GASAR
- [ ] Innoviva - Surveillance of global clinical isloates of Acinetobacter baumannii-calco-aceticus complex
- [x] Shionogi - SIDERO-WT
- [x] Global AMR R&D Hub (Required for eligibility for the Global AMR R&D Hub Cross-Domain award)
- [x] GRAM 2019 Lancet burden estimates — https://doi.org/10.1016/S0140-6736(21)02724-0, EUCAST Clinical Breakpoint Tables v14.0 (2024) — https://www.eucast.org

6. Did your team use any generative AI tools (e.g., ChatGPT, Claude, Copilot, Gemini) during this challenge — for tasks such as data analysis, visualization, writing, or code generation? *

Use of generative AI tools does not disqualify a team from the challenge. We simply ask for honest and complete disclosure.

- (selected) Yes
- No

7. For the usage of generative AI, please describe: which tools were used, what tasks they assisted with, and how central their contribution was to your final product (e.g., minor assistance vs. substantially shaped the analysis or narrative). *

Tool used: Claude (Anthropic), primarily through the claude.ai web interface,
Tasks the tool assisted with as detailed below (i) helped iteratively debug the Python codebase (data harmonisation across three surveillance datasets); the tool helped reviewed the code drafts; the human team reviewed, tested, decided when to deviate, and verified every output. (ii) helped stress-test the original causal framing (funding-suppresses-resistance) and then helped identified potential compliance flags on section naming that we missed.

Tasks that were NOT delegated to the tool are listed as follow (i) Verification of primary sources: the human team verified EUCAST breakpoints line-by-line against the official v14.0 PDF, verified GRAM burden estimates against Table S22 of the Lancet appendix, and cross-checked totals against the paper's published figures. Wrong values from the tool's earlier searches were caught by human verification. (ii) EUCAST breakpoint classification engine, latent class analysis for resistance phenotyping via stepmix, Funding Allocation Profile construction from Hub Categories, Global Misalignment Index computation, Bayesian random-walk Beta-binomial projection in Stan, projected GMI with uncertainty propagation, pipeline cross-flag with quadrant assignment, figure generation via matplotlib). (iii) the clinical decision to fix Klebsiella LCA at K=5 (rather than the BIC-preferred K=6) was made by the team based on clinical interpretability. The scope decision to project only K. pneumoniae and A. baumannii was made by the team. (iv) all data downloads, code execution, and file uploads were performed by the team on our own machines. (v) The substantial code for pipeline implementation , report structure, and figure production; central (but-not-sole) for analytical design (framings were proposed by us and iterated also; nfor scientific verification (all values checked against primary sources by us) and no domain judgment calls were delegated. The team retained full editorial control at every stage.

8. How did your team hear about the Vivli AMR Data Challenge? *

- [x] ESCMID Global 2026, Munich, Germany
- [ ] IMARI 2026, Las Vegas, NV, United States
- [x] The Global Health Network Communications/Events
- [x] Global AMR R&D Hub communications or events
- [x] From a past data challenge awardee or participant
- [ ] Vivli newsletter or email list
- [ ] Colleague or advisor recommendation
- [ ] Social media
- [ ] Other

9. What is your team's primary motivation for entering the 2026 Vivli AMR Surveillance Data Challenge? *